1. Executive Summary

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1. Executive summary
- Train games=940 | Derby games=75
- Nested CV=5 outer folds x 3 inner folds (expanding time-aware)
- Selected model family: Regime-aware simplex stack (Ridge/Huber/HistGB/GBR/HistGB-bag) + q50 blend
- Selected Elo variant: elo_base_static
- Nested outer RMSE/MAE: 43.563 / 34.499
- Selected postprocess module/eta: expand_affine_global / 0.55
- Calibration=pooled | Scale=none | Regime stack=False
- Runtime safety: fast_mode=False | budget_triggered=histgb_bag_auto_reduce, max_fits
```

1B. Runtime Bottleneck And Fixes

Runtime bottleneck and fixes
- Profile-first instrumentation added for hotspot phase totals, per-family _predict_family timing, model fit counts, and top slow events.
- Shared split-level matrix/weight cache keyed by train/val index signatures + feature signature + half-life to eliminate repeated _xy/time-weight prep.
- Bundled family prediction path prepares X/y/w once per split and fits Ridge/Huber/HistGB/GBR/HistGB-bag/q50 sequentially.
- HistGB bagging reduced to 2 models by default (fast mode uses 1), with deterministic seed schedule and budget-aware auto-reduction.
- Candidate scan uses conservative early pruning: coarse score on first inner split, then full evaluation on top candidates only.
- Budget enforcement: total time, scan time, and fit cap stop further scanning/training and continue with best-so-far.
- Budgets triggered: histgb_bag_auto_reduce, max_fits
- Fast mode=False; optimization toggles enabled=True.
- Reproducibility: seed=23, OMP_NUM_THREADS=, MKL_NUM_THREADS=.
- Artifact validation: predictions rows=75 (expected 75), rankings rows=165 (expected 165), ranks unique 1..165=True.

1C. Fast Profiling Before vs After (timing / fits) (1/1)

kind	name	before_sec	after_sec	delta_sec
phase	_scan_and_select_core_candidates	5.230	5.524	0.294
phase	_core_scan_score	5.226	5.516	0.290
phase	_generate_inner_oof_and_outer_preds	5.540	3.973	-1.567
phase	_predict_family	10.500	9.416	-1.084
phase	_build_split_tables	0.447	0.334	-0.113
phase	_prepare_variant_outer_data	0.550	0.440	-0.110
fit_count	histgb	48.000	51.000	3.000
fit_count	histgb_bag	8.000	6.000	-2.000
fit_count	histgb_q50	8.000	6.000	-2.000
fit_count	huber	8.000	6.000	-2.000
fit_count	ridge	48.000	51.000	3.000

1D. Runtime Focus Phase Timing Summary (1/1)

phase	total_sec	calls
_predict_family	211.260	10520
_scan_and_select_core_candidates	137.848	6
_core_scan_score	137.763	3480
_generate_inner_oof_and_outer_preds	78.755	48
_prepare_variant_outer_data	5.270	24
_build_split_tables	4.753	96

1E. _predict_family Aggregated Timing / Fit Counts (1/1)

family	predict_time_sec	predict_calls	fit_count
histgb	139.886	4873	1170
histgb_q50	23.931	193	32
gbr	20.922	193	32
histgb_bag	17.403	193	64
huber	7.398	193	32
ridge	1.720	4873	1170
histgb_q80	0.000	1	0
histgb_q20	0.000	1	0

2-4. Changes / Leakage / Nested CV

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2. What changed vs previous pipeline
- Nested tuning for post-model layers and key hyperparameters (Ridge alpha, HistGB grid, recency half-life).
- Regime-specific simplex stacking + regime-specific calibration + regime/pool scale correction with fallback.
- Recency features: EMA multi-alpha, last3/5/8, trend, consistency ratio.
- Upgraded Elo variants: dynamic-K, team-specific regularized home effect, inactivity decay.
- HistGB seed bagging and quantile HistGB (median, plus q20/q80 for diagnostics).
- Shift diagnostics (KS + PSI + out-of-range fraction) and OOD-triggered mitigation.

3. Data constraints and leakage controls
- All features are pregame-only and built sequentially.
- Static rating models refit inside each train split; no outer-fold leakage in postprocessing.
- Derby predictions are neutral-site (home effects zeroed for derby features).

4. Nested CV design (outer/inner schematic)
 outer_fold=1 train_n=125 val_n=193 train_end=2025-01-12 val=2025-01-23..2025-02-19
 outer_fold=2 train_n=318 val_n=205 train_end=2025-02-19 val=2025-03-03..2025-04-02
 outer_fold=3 train_n=523 val_n=151 train_end=2025-04-02 val=2025-04-14..2025-04-30
 outer_fold=4 train_n=674 val_n=102 train_end=2025-04-30 val=2025-05-13..2025-05-29
 outer_fold=5 train_n=776 val_n=164 train_end=2025-05-29 val=2025-05-30..2025-06-30
```

5. Outer-fold selections and metrics (1/1)

outer_candidate_id	dtype	feature_set	n_features	lifted	edge_lifted	interbond_stop	pule_mean	cap_mean	mean_pos	mean_neg	tail_distance	cluster_prop	prop_mean	mean_mean	mean_q50	outer_n	outer_roc_auc	outer_pr_auc	outer_tail_disp	outer_tail_cov	outer_cov_gap	outer_ece	outer_brier	outer_nll	outer_acc	outer_bal_acc	outer_f1	outer_mcc	outer_sens	outer_spec	outer_prec	outer_npv	outer_fpr	outer_fnr	outer_pos_rate	outer_pred_rate	outer_lift	outer_ap_gain	outer_ap_ratio	outer_scale_q50	outer_b_hom	outer_s_cap_
fold_4_regime_2_stack_0_regime	regime	affine	2	True	True	True	0.556	3.62	4.57	1.83	0.355	5.25	0.36	0.000	True	42.34	45.81	47.30	6.64	8.80	3.66	0.53	3.59	0.388	8.11	45.09	0.84	4.55	2.75	True	pooled	none	0.000	0.010	False
expand_affine_total_stack_2_001_expand_affine	affine	total	2	False	False	False	0.556	5.27	5.74	0.50	0.036	7.35	0.025	0.000	False	45.32	45.32	44.36	4.89	3.32	7.61	0.00	1.86	0.00	0.00	44.24	0.00	4.36	0.00	False	pooled	none	0.000	0.000	True
expand_affine_total_stack_3_001_expand_affine	affine	total	3	False	False	False	0.556	4.81	5.38	9.77	0.07	7.33	0.025	0.000	True	44.41	44.41	38.46	0.91	1.88	6.17	0.01	4.24	0.317	0.56	38.59	0.01	8.59	0.133	True	pooled	pooled	0.000	0.000	True
expand_affine_total_stack_2_001_expand_affine	affine	total	2	False	False	False	0.556	3.17	4.18	6.52	0.04	9.05	0.003	0.000	False	43.14	43.14	40.49	2.32	0.24	7.03	0.00	0.46	0.00	0.00	40.49	0.00	0.49	0.000	False	pooled	none	0.000	0.000	True
expand_affine_total_stack_3_001_expand_affine	affine	total	3	False	False	False	0.556	2.16	3.63	2.87	0.05	8.56	0.03	0.000	False	41.96	41.96	47.18	7.72	2.14	3.46	0.00	1.16	0.00	0.00	47.13	0.00	7.18	0.000	False	pooled	none	0.000	0.000	True

5. Outer-fold RMSE/MAE table (1/1)

outer_fold	rmse	mae	bias	pred_std	actual_std	resid_std	tail_rmse	xtail_rmse	tail_bias	tail_dispersion_rat	corr
1	47.308	36.643	-8.806	38.119	45.098	46.481	63.662	79.784	-13.594	0.535	0.386
2	44.366	34.892	-3.325	0.000	44.241	44.241	77.618	86.835	-21.862	0.000	0.000
3	38.463	30.913	1.883	0.562	38.591	38.417	66.175	71.517	4.241	0.010	0.317
4	40.497	32.322	0.247	0.000	40.496	40.496	70.038	78.220	-10.463	0.000	0.000
5	47.182	37.726	-2.143	0.000	47.133	47.133	83.461	92.229	-11.165	0.000	0.000

6. Feature additions and recency ablations (core scan) (1/1)

feature_profile	half_life_days	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
compact_recency	60.000	42.599	45.534	44.449	483
compact_recency		42.608	45.539	44.462	486
compact_recency	30.000	42.625	45.542	44.513	487
full_recency	30.000	42.672	45.579	44.590	502
full_recency		42.680	45.586	44.600	502
full_recency	60.000	42.923	45.945	44.817	440

6. Core scan results (top rows) (1/4)

candidate_ke	elo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber	scan_ridge_s	scan_stage	pred_std_sca	scan_splits_u	scan_stage	outer_fold
recency|hel	elo_base_sta	full_recenc		16.000	3	38.259	40.520	32.612					41.458	33.976	-29.249	1	coarse	1
recency|hel	elo_dynamic	full_recenc		16.000	3	38.259	40.520	32.612					41.458	33.976	-29.249	1	coarse	1
full_recen	elo_dynamic_k_	full_recenc		16.000	3	38.259	40.520	32.612					41.458	33.976	-29.249	1	coarse	1
cay|full_recen	elo_dynamic_k_team	full_recenc		16.000	3	38.259	40.520	32.612					41.458	33.976	-29.249	1	coarse	1
l_recency|hel	elo_base_sta	full_recenc	30.000	16.000	3	38.263	40.519	32.619					41.462	33.978	-29.308	1	coarse	1
_recency|hel	elo_dynamic	full_recenc	30.000	16.000	3	38.263	40.519	32.619					41.462	33.978	-29.308	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	30.000	16.000	3	38.263	40.519	32.619					41.462	33.978	-29.308	1	coarse	1
ecay|full_recen	elo_dynamic_k_team	full_recenc	30.000	16.000	3	38.263	40.519	32.619					41.462	33.978	-29.308	1	coarse	1
l_recency|hel	elo_base_sta	full_recenc	60.000	16.000	2	38.266	40.602	32.942					41.460	33.977	-28.477	1	coarse	1
_recency|hel	elo_dynamic	full_recenc	60.000	16.000	2	38.266	40.602	32.942					41.460	33.977	-28.477	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	60.000	16.000	2	38.266	40.602	32.942					41.460	33.977	-28.477	1	coarse	1
ecay|full_recen	elo_dynamic_k_team	full_recenc	60.000	16.000	2	38.266	40.602	32.942					41.460	33.977	-28.477	1	coarse	1
l_recency|hel	elo_base_sta	full_recenc	60.000	16.000	3	38.272	40.538	32.625					41.460	33.977	-29.304	1	coarse	1
_recency|hel	elo_dynamic	full_recenc	60.000	16.000	3	38.272	40.538	32.625					41.460	33.977	-29.304	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	60.000	16.000	3	38.272	40.538	32.625					41.460	33.977	-29.304	1	coarse	1
ecay|full_recen	elo_dynamic_k_team	full_recenc	60.000	16.000	3	38.272	40.538	32.625					41.460	33.977	-29.304	1	coarse	1
l_recency|hel	elo_base_sta	full_recenc	30.000	16.000	2	38.272	40.611	32.956					41.462	33.978	-28.494	1	coarse	1
_recency|hel	elo_dynamic	full_recenc	30.000	16.000	2	38.272	40.611	32.956					41.462	33.978	-28.494	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	30.000	16.000	2	38.272	40.611	32.956					41.462	33.978	-28.494	1	coarse	1
ecay|full_recen	elo_dynamic_k_team	full_recenc	30.000	16.000	2	38.272	40.611	32.956					41.462	33.978	-28.494	1	coarse	1
recency|hel	elo_base_sta	full_recenc		16.000	2	38.279	40.627	32.984					41.458	33.976	-28.480	1	coarse	1
recency|hel	elo_dynamic	full_recenc		16.000	2	38.279	40.627	32.984					41.458	33.976	-28.480	1	coarse	1
full_recen	elo_dynamic_k_	full_recenc		16.000	2	38.279	40.627	32.984					41.458	33.976	-28.480	1	coarse	1
cay|full_recen	elo_dynamic_k_team	full_recenc		16.000	2	38.279	40.627	32.984					41.458	33.976	-28.480	1	coarse	1

6. Core scan results (top rows) (2/4)

candidate_ke	elo_variat	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber	scan_ridge_s	scan_stage	pred_std_sca	scan_splits_u	scan_stage	outer_fold
ct_recency	elo_basesta	compact_rece		16.000	3	38.309	40.520	32.612					41.600	34.112	-29.249	1	coarse	1
ct_recency	elo_dynamic	compact_rece		16.000	3	38.309	40.520	32.612					41.600	34.112	-29.249	1	coarse	1
mpact_rece	elo_dynamic_d	compact_rece		16.000	3	38.309	40.520	32.612					41.600	34.112	-29.249	1	coarse	1
y|cand	elo_dynamic_k_team	compact_rece		16.000	3	38.309	40.520	32.612					41.600	34.112	-29.249	1	coarse	1
act_recency	elo_basesta	compact_rece	30.000	16.000	3	38.310	40.519	32.619					41.598	34.110	-29.308	1	coarse	1
act_recency	elo_dynamic	compact_rece	30.000	16.000	3	38.310	40.519	32.619					41.598	34.110	-29.308	1	coarse	1
ompact_rece	elo_dynamic_d	compact_rece	30.000	16.000	3	38.310	40.519	32.619					41.598	34.110	-29.308	1	coarse	1
y|cand	elo_dynamic_k_team	compact_rece	30.000	16.000	3	38.310	40.519	32.619					41.598	34.110	-29.308	1	coarse	1
act_recency	elo_basesta	compact_rece	60.000	16.000	2	38.315	40.602	32.942					41.599	34.111	-28.477	1	coarse	1
act_recency	elo_dynamic	compact_rece	60.000	16.000	2	38.315	40.602	32.942					41.599	34.111	-28.477	1	coarse	1
ompact_rece	elo_dynamic_d	compact_rece	60.000	16.000	2	38.315	40.602	32.942					41.599	34.111	-28.477	1	coarse	1
y|cand	elo_dynamic_k_team	compact_rece	60.000	16.000	2	38.315	40.602	32.942					41.599	34.111	-28.477	1	coarse	1
act_recency	elo_basesta	compact_rece	30.000	16.000	2	38.320	40.611	32.956					41.598	34.110	-28.494	1	coarse	1
act_recency	elo_dynamic	compact_rece	30.000	16.000	2	38.320	40.611	32.956					41.598	34.110	-28.494	1	coarse	1
ompact_rece	elo_dynamic_d	compact_rece	30.000	16.000	2	38.320	40.611	32.956					41.598	34.110	-28.494	1	coarse	1
y|cand	elo_dynamic_k_team	compact_rece	30.000	16.000	2	38.320	40.611	32.956					41.598	34.110	-28.494	1	coarse	1
act_recency	elo_basesta	compact_rece	60.000	16.000	3	38.321	40.538	32.625					41.599	34.111	-29.304	1	coarse	1
act_recency	elo_dynamic	compact_rece	60.000	16.000	3	38.321	40.538	32.625					41.599	34.111	-29.304	1	coarse	1
ompact_rece	elo_dynamic_d	compact_rece	60.000	16.000	3	38.321	40.538	32.625					41.599	34.111	-29.304	1	coarse	1
y|cand	elo_dynamic_k_team	compact_rece	60.000	16.000	3	38.321	40.538	32.625					41.599	34.111	-29.304	1	coarse	1
ct_recency	elo_basesta	compact_rece		16.000	2	38.329	40.627	32.984					41.600	34.112	-28.480	1	coarse	1
ct_recency	elo_dynamic	compact_rece		16.000	2	38.329	40.627	32.984					41.600	34.112	-28.480	1	coarse	1
mpact_rece	elo_dynamic_d	compact_rece		16.000	2	38.329	40.627	32.984					41.600	34.112	-28.480	1	coarse	1
y|cand	elo_dynamic_k_team	compact_rece		16.000	2	38.329	40.627	32.984					41.600	34.112	-28.480	1	coarse	1

6. Core scan results (top rows) (3/4)

candidate_ke	elo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_u	scan_gbr_sc	scan_gbr_u	scan_huber	scan_huber_u	scan_ridge_s	scan_ridge_u	scaled_std_sc	scan_splits_u	scan_stage	outer_fold
_recency|ele	elo_base_sta	full_recenc		8.000	3	38.336	40.520	32.612					41.678	34.139	-29.249	1	coarse	1
_recency|ele	elo_dynamic	full_recenc		8.000	3	38.336	40.520	32.612					41.678	34.139	-29.249	1	coarse	1
|full_recen	elo_dynamic_k_	full_recenc		8.000	3	38.336	40.520	32.612					41.678	34.139	-29.249	1	coarse	1
cay|full_recen	elo_dynamic_k_tea	full_recenc		8.000	3	38.336	40.520	32.612					41.678	34.139	-29.249	1	coarse	1
ll_recency|ele	elo_base_sta	full_recenc	30.000	8.000	3	38.337	40.519	32.619					41.675	34.144	-29.308	1	coarse	1
l_recency|ele	elo_dynamic	full_recenc	30.000	8.000	3	38.337	40.519	32.619					41.675	34.144	-29.308	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	30.000	8.000	3	38.337	40.519	32.619					41.675	34.144	-29.308	1	coarse	1
ecay|full_recen	elo_dynamic_k_tea	full_recenc	30.000	8.000	3	38.337	40.519	32.619					41.675	34.144	-29.308	1	coarse	1
ll_recency|ele	elo_base_sta	full_recenc	60.000	8.000	2	38.342	40.602	32.942					41.676	34.141	-28.477	1	coarse	1
l_recency|ele	elo_dynamic	full_recenc	60.000	8.000	2	38.342	40.602	32.942					41.676	34.141	-28.477	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	60.000	8.000	2	38.342	40.602	32.942					41.676	34.141	-28.477	1	coarse	1
ecay|full_recen	elo_dynamic_k_tea	full_recenc	60.000	8.000	2	38.342	40.602	32.942					41.676	34.141	-28.477	1	coarse	1
ll_recency|ele	elo_base_sta	full_recenc	30.000	8.000	2	38.347	40.611	32.956					41.675	34.144	-28.494	1	coarse	1
l_recency|ele	elo_dynamic	full_recenc	30.000	8.000	2	38.347	40.611	32.956					41.675	34.144	-28.494	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	30.000	8.000	2	38.347	40.611	32.956					41.675	34.144	-28.494	1	coarse	1
ecay|full_recen	elo_dynamic_k_tea	full_recenc	30.000	8.000	2	38.347	40.611	32.956					41.675	34.144	-28.494	1	coarse	1
ll_recency|ele	elo_base_sta	full_recenc	60.000	8.000	3	38.348	40.538	32.625					41.676	34.141	-29.304	1	coarse	1
l_recency|ele	elo_dynamic	full_recenc	60.000	8.000	3	38.348	40.538	32.625					41.676	34.141	-29.304	1	coarse	1
a|full_recen	elo_dynamic_k_	full_recenc	60.000	8.000	3	38.348	40.538	32.625					41.676	34.141	-29.304	1	coarse	1
ecay|full_recen	elo_dynamic_k_tea	full_recenc	60.000	8.000	3	38.348	40.538	32.625					41.676	34.141	-29.304	1	coarse	1
_recency|ele	elo_base_sta	full_recenc		8.000	2	38.357	40.627	32.984					41.678	34.139	-28.480	1	coarse	1
_recency|ele	elo_dynamic	full_recenc		8.000	2	38.357	40.627	32.984					41.678	34.139	-28.480	1	coarse	1
|full_recen	elo_dynamic_k_	full_recenc		8.000	2	38.357	40.627	32.984					41.678	34.139	-28.480	1	coarse	1
cay|full_recen	elo_dynamic_k_tea	full_recenc		8.000	2	38.357	40.627	32.984					41.678	34.139	-28.480	1	coarse	1

6. Core scan results (top rows) (4/4)

candidate_k	elo_variant	feature_prof	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber	scan_ridge_s	scan_ridge_	scan_splits_u	scan_stage	outer_fold
ll_recency|elo	elo_base_sta	full_recenc	30.000	4.000	3	38.438	40.519	32.619					41.964	34.084	-29.308	1	coarse	1
l_recency|elo	elo_dynamic	full_recenc	30.000	4.000	3	38.438	40.519	32.619					41.964	34.084	-29.308	1	coarse	1
a|full_recency	elo_dynamic_k_f	full_recenc	30.000	4.000	3	38.438	40.519	32.619					41.964	34.084	-29.308	1	coarse	1
ecency|full_recen	elo_dynamic_k_team	full_recenc	30.000	4.000	3	38.438	40.519	32.619					41.964	34.084	-29.308	1	coarse	1
_recency|elo	elo_base_sta	full_recenc		4.000	3	38.439	40.520	32.612					41.973	34.078	-29.249	1	coarse	1
_recency|elo	elo_dynamic	full_recenc		4.000	3	38.439	40.520	32.612					41.973	34.078	-29.249	1	coarse	1
|full_recency	elo_dynamic_k_f	full_recenc		4.000	3	38.439	40.520	32.612					41.973	34.078	-29.249	1	coarse	1
cay|full_recen	elo_dynamic_k_team	full_recenc		4.000	3	38.439	40.520	32.612					41.973	34.078	-29.249	1	coarse	1

7. Elo variant comparisons (1/1)

elo_variant	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
elo_base_static	42.257	45.001	44.239	959
elo_dynamic_k_teamha_decay	42.876	45.846	44.499	625
elo_dynamic_k_teamha	42.897	45.953	44.839	657
elo_dynamic_k	42.897	45.954	44.844	659

7. HistGB hyperparameter sweep summary (1/1)

histgb_idx	scan_score_mean	scan_histgb_rmse_mean	n	params
3	41.996	44.203	500	{'learning_rate': 0.025, 'max_depth': 5, 'max_leaf_nodes': 63, 'min_samples_leaf': 5
2	41.975	44.205	500	{'learning_rate': 0.03, 'max_depth': 4, 'max_leaf_nodes': 63, 'min_samples_leaf': 6,
4	42.166	44.566	500	{'learning_rate': 0.06, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 12
1	42.651	45.713	500	{'learning_rate': 0.05, 'max_depth': 4, 'max_leaf_nodes': 31, 'min_samples_leaf': 8,
0	44.805	49.772	500	{'learning_rate': 0.04, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10

8. Regime-specific stacking / calibration results (1/1)

module_name	use_regime_sta	calibration_mod	scale_mode	eta	outer_rmse_mo	outer_mae_mo	outer_tail_dispersion	outer_tail_bias_m	outer_corr_mo	outer_dispersion_	regime_mi	gain_vs_poole	n
regime_affine_v	True	pooled	pooled	0.550	38.463	30.913	66.175	0.010	4.241	0.317	0.015	0.133	8
xpand_affine_gl	False	pooled	none	0.550	44.123	35.032	76.049	0.051	-14.752	0.035	0.066	0.000	26
and_regime_affi	True	pooled	none	0.550	48.062	36.980	64.718	0.559	-13.998	0.379	0.866	-3.379	6

9. Dispersion and scale correction analysis (1/1)

outer_fold	stage	pred_std	actual_std	rmse
1	raw_mean	25.955	45.098	43.383
1	calibrated	32.883	45.098	45.244
1	scaled	32.883	45.098	45.244
1	final	38.119	45.098	47.308
2	raw_mean	0.000	44.241	44.274
2	calibrated	0.000	44.241	44.268
2	scaled	0.000	44.241	44.268
2	final	0.000	44.241	44.366
3	raw_mean	0.000	38.591	38.597
3	calibrated	0.000	38.591	39.006
3	scaled	0.000	38.591	38.662
3	final	0.562	38.591	38.463
4	raw_mean	0.000	40.496	40.528
4	calibrated	0.000	40.496	40.576
4	scaled	0.000	40.496	40.576
4	final	0.000	40.496	40.497
5	raw_mean	0.000	47.133	47.137
5	calibrated	0.000	47.133	47.557
5	scaled	0.000	47.133	47.557
5	final	0.000	47.133	47.182

10. Shift diagnostics: Train recent vs Derby (1/1)

feature	train_recent_mean	derby_mean	train_recent_std	derby_std	ks_stat	ks_pvalue	psi	derby_outside_train_1_9	ood_flag
games_played_min	9.122	11.067	1.095	0.660	0.730	0.000	8.252	0.227	True
elo_diff_pre	58.187	-0.325	64.506	58.625	0.409	0.000	2.070	0.107	True
same_conf_flag	0.697	0.013	0.459	0.115	0.684	0.000	0.000	0.000	True
massey_diff	5.872	2.195	19.548	15.187	0.166	0.075	0.192	0.000	False
consistency_ratio_di	0.059	-0.008	0.657	0.577	0.113	0.427	0.167	0.013	False
_margin_last5_vs_sea	-1.434	1.554	16.927	19.606	0.105	0.514	0.133	0.067	False
volatility_sum	71.164	73.217	11.169	9.954	0.139	0.196	0.132	0.027	False
offdef_net_diff	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False
offdef_margin_neutr	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False
elo_neutral_diff_pre	3.187	-0.325	64.506	58.625	0.099	0.591	0.053	0.013	False

10. Shift Mitigation Notes

10. Shift mitigation rules
- Detected 3 OOD-like feature shifts (KS/PSI/out-of-range checks).
- Applied softer scaling (blend toward 1.0) and wider winsor bounds under OOD risk.

8. Regime weights (outer_fold_1_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	103	0.301	0.350	0.000	0.350
full	moderate|cross|lowinfo|highv	68	0.601	0.199	0.000	0.199
simple	moderate|cross|lowinfo	68	0.601	0.199	0.000	0.199
gap	moderate	93	0.710	0.145	0.000	0.145

8. Regime weights (outer_fold_2_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	284	0.250	0.250	0.250	0.250

8. Regime weights (outer_fold_3_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	448	0.250	0.250	0.250	0.250
full	moderate|cross|lowinfo|highv	70	0.250	0.250	0.250	0.250
full	moderate|same|lowinfo|highv	67	0.250	0.250	0.250	0.250
full	moderate|same|lowinfo|lowv	88	0.250	0.250	0.250	0.250
simple	moderate|cross|lowinfo	96	0.250	0.250	0.250	0.250
simple	moderate|same|lowinfo	155	0.250	0.250	0.250	0.250
simple	close|same|lowinfo	62	0.250	0.250	0.250	0.250
gap	moderate	282	0.250	0.250	0.250	0.250
gap	close	107	0.250	0.250	0.250	0.250
gap	mismatch	59	0.250	0.250	0.250	0.250

8. Regime weights (outer_fold_4_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	549	0.250	0.250	0.250	0.250

8. Regime weights (outer_fold_5_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	589	0.250	0.250	0.250	0.250

8. Calibration params (outer_fold_1_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	103	1.152	1.267

8. Calibration params (outer_fold_2_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	284	-3.686	1.063

8. Calibration params (outer_fold_3_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	448	9.285	0.300

8. Calibration params (outer_fold_4_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	549	5.098	0.300

8. Calibration params (outer_fold_5_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	589	11.008	0.300

9. Scale params (outer_fold_1_selected) (1/1)

level	regime	n	scale
pooled	ALL	103	1.000

9. Scale params (outer_fold_2_selected) (1/1)

level	regime	n	scale
pooled	ALL	284	1.000

9. Scale params (outer_fold_3_selected) (1/1)

level	regime	n	scale
pooled	ALL	448	0.700

9. Scale params (outer_fold_4_selected) (1/1)

level	regime	n	scale
pooled	ALL	549	0.700

9. Scale params (outer_fold_5_selected) (1/1)

level	regime	n	scale
pooled	ALL	589	0.700

11. Residual Diagnostics (Summary)

```
11. Residual diagnostics
- OOF final RMSE=44.186, MAE=34.818, Bias=-2.973
- Residual std=44.086, skew=0.008, kurtosis=-0.001
- Mean-stack RMSE=43.214
- Quantile median (q50) RMSE=42.315
- Mean+median blend RMSE=43.214
- Calibrated RMSE=43.826
- Tail error threshold (top 10%)=71.753
```

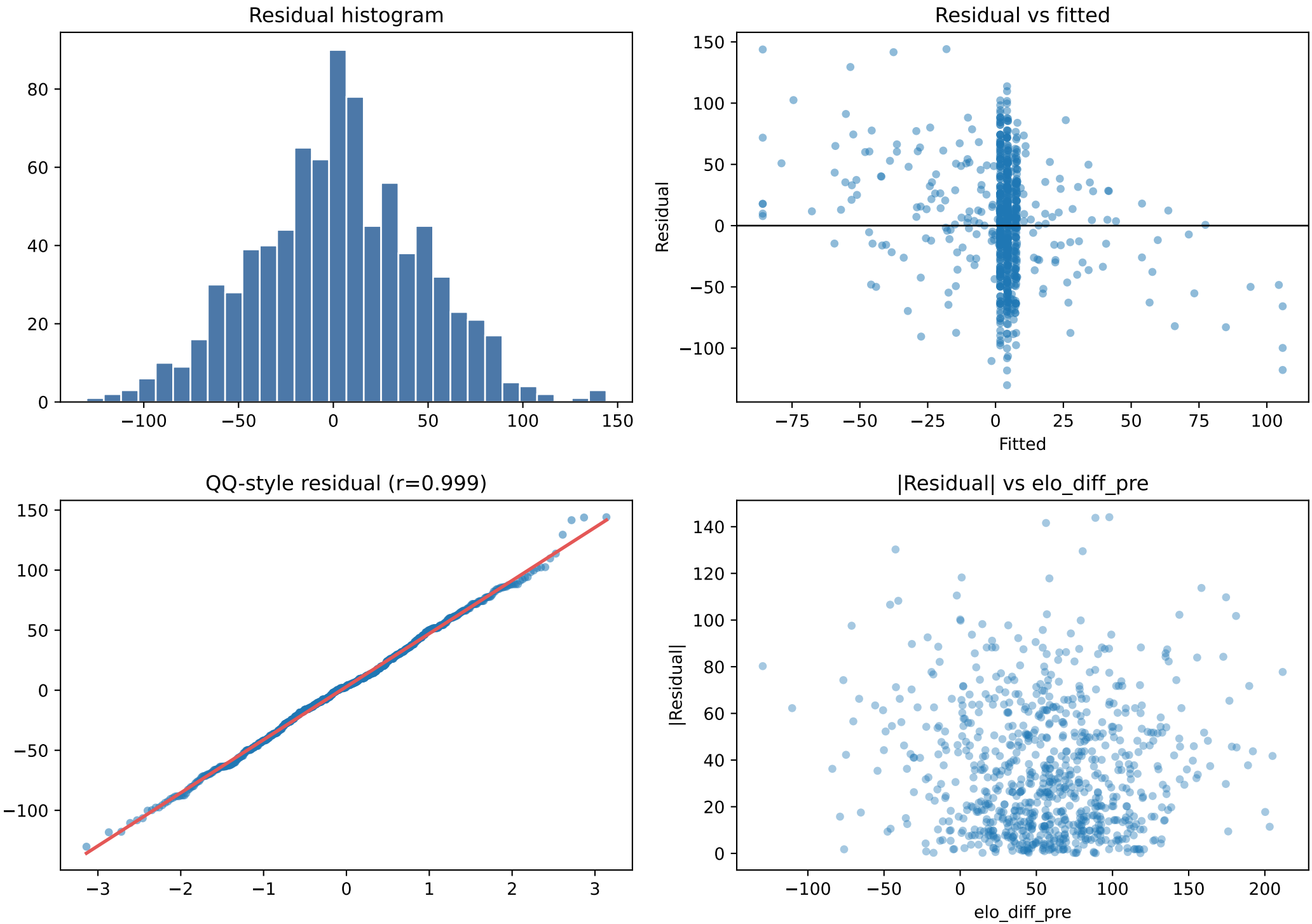
11. Tail error diagnostics (top 10% abs-error) (1/2)

Date	ado_diff	passes	off	efficiency	volatility	daily_sales	avg_sales	payment	payment_count	month	month_sin	month_cos	consistency	row_index	true	pred_raw	pred_hist	pred_cal	hist	pred_q	outer	fold	raw	raw_cal	pred_scaled	pred_final	model	abs_error
-02-19	0	97.907	12.251	6.286	33.893	31.000	4.000	2.000	1.000	0.000	0.000	0.000	1.510	269	126.00	56.541	63.719	9.888	8.862	9.888	0.968	module=3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	144.073
-01-24	0	38.771	19.353	17.107	45.000	3.000	1.000	2.000	1.000	0.000	0.000	-16.126	178	58.00	111.82	143.75	35.230	46.304	35.230	41.800	module=6	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	143.80
-02-19	0	56.345	0.396	12.250	23.795	26.949	3.000	3.000	1.000	0.000	0.000	0.000	0.499	304	104.00	17.526	67.599	32.213	31.243	32.213	30.711	module=2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	141.59
-05-16	0	-42.443	28.509	42.000	34.461	34.372	10.000	10.000	1.000	0.000	15.400	16.480	-0.336	899	-126.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	130.24
-02-05	0	30.416	6.227	-0.857	40.000	40.000	2.000	2.000	0.000	1.000	0.000	0.000	0.829	221	76.00	42.924	96.543	13.791	16.905	13.791	10.244	module=1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	129.49
-05-30	0	0.973	-15.184	19.896	54.529	26.609	11.000	10.000	0.000	1.000	-3.582	-3.755	0.377	829	-114.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	118.24
-02-05	0	58.564	3.601	-4.571	10.000	3.000	2.000	2.000	1.000	0.000	0.000	0.000	43.250	222	-12.00	143.75	163.71	17.833	23.755	17.833	24.495	module=1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	117.84
-05-16	0	58.382	2.158	21.600	60.669	23.457	10.000	10.000	1.000	0.000	38.800	38.931	-1.679	903	118.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	113.75
-02-19	0	-2.217	-0.798	-8.000	10.625	39.708	3.000	4.000	1.000	0.000	0.000	0.000	0.717	267	-112.00	97.144	12.254	42.034	47.094	42.034	21.188	module=1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	110.51
-05-16	0	74.545	36.803	38.029	37.250	48.077	9.000	10.000	1.000	0.000	-10.800	11.661	-0.041	890	114.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	109.75
-05-16	0	-40.628	14.846	26.282	4.985	41.395	11.000	11.000	0.000	1.000	11.745	13.361	1.457	907	-104.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	108.24
-05-14	0	-46.000	28.029	35.306	53.343	32.450	10.000	8.000	0.000	1.000	37.000	37.984	-0.010	687	-102.00	5.953	5.953	5.953	5.953	5.953	5.953	module=4	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	106.58
-02-19	0	56.923	-1.617	10.851	18.833	70.370	3.000	3.000	1.000	0.000	0.000	0.000	0.022	278	28.00	45.566	64.986	39.724	44.064	39.724	30.344	module=1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	102.47
-04-02	0	43.902	1.036	19.782	43.916	34.006	5.000	7.000	1.000	0.000	-0.857	-1.214	0.386	488	104.00	6.767	6.767	6.767	6.767	6.767	6.767	module=2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	102.26
-05-12	0	81.092	8.105	26.152	7.697	31.404	8.000	10.000	1.000	0.000	-8.500	10.079	-0.433	851	106.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	101.75
-05-27	0	0.018	-4.515	13.129	9.976	42.180	10.000	11.000	0.000	1.000	-18.695	-16.859	-0.354	928	-96.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	100.24
-01-24	0	79.127	6.934	-2.500	0.000	56.000	2.000	2.000	0.000	1.000	0.000	0.000	35.313	155	6.000	143.75	163.71	-2.129	-9.354	-2.129	-8.362	module=1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	99.84
-05-30	0	0.257	-6.552	16.219	2.571	29.441	8.000	9.000	1.000	0.000	-12.067	-11.931	1.381	804	104.00	5.740	5.740	5.740	5.740	5.740	5.740	module=5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	99.75
-03-04	0	14.546	5.403	-0.022	35.281	25.020	4.000	4.000	1.000	0.000	0.000	0.000	-0.107	385	100.00	6.767	6.767	6.767	6.767	6.767	6.767	module=2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	98.26
-03-03	0	31.552	0.049	-5.200	47.864	28.914	4.000	4.000	1.000	0.000	0.000	0.000	-0.165	325	-96.00	6.767	6.767	6.767	6.767	6.767	6.767	module=2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	97.73
-04-30	0	71.266	26.495	-36.000	30.786	33.790	8.000	8.000	1.000	0.000	11.000	11.683	1.659	621	-90.00	6.099	6.099	6.099	6.099	6.099	6.099	module=3	expan	[illegible]	[illegible]	[illegible]	[illegible]	97.56
-04-02	0	54.229	5.403	-0.800	62.007	30.529	4.000	6.000	1.000	0.000	11.200	11.185	-0.191	491	-94.00	6.767	6.767	6.767	6.767	6.767	6.767	module=2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	95.73

11. Tail error diagnostics (top 10% abs-error) (2/2)

Date	ado_diff	massoff	def_vol	utility	volatility	daily_spread	pct_spread	bank_rate	dow	month	month_end	month_start	pct_5	pct_95	consistency	row_index	y_true	used_prior	used_hurst	used_hist	used_reg	used_hist	used_reg	used_cluster	used_fd	used_raw	used_raw	used_raw	used_calibrated	used_scaled	used_fallback	used_post	used_post	used_post	abs_error
-03-04	072.641	24.183	26.626	46.106	34.500	5.000	3.000	1.000	0.000	0.000	0.000	0.000	0.923	365	96.000	6.767	6.767	6.767	6.767	6.767	6.767	module=	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	model=	pool=	4.262
-05-27	099.197	18.635	16.943	24.687	37.918	10.000	10.000	1.000	0.000	14.000	13.399	-0.166	916	98.000	5.740	5.740	5.740	5.740	5.740	5.740	module=	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	model=	pool=	3.753
-03-04	07.705	-6.828	12.221	8.219	11.776	3.000	3.000	1.000	0.000	0.000	0.000	3.604	342	-92.000	6.767	6.767	6.767	6.767	6.767	6.767	module=	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	model=	pool=	3.738
-05-15	021.366	26.736	33.143	45.584	44.463	8.000	8.000	0.000	1.000	-43.204	-43.019	0.061	697	-88.000	5.953	5.953	5.953	5.953	5.953	5.953	module=	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	model=	pool=	2.580
-04-02	038.114	-4.612	12.442	1.784	22.464	5.000	5.000	0.000	1.000	0.000	0.000	1.407	495	94.000	6.767	6.767	6.767	6.767	6.767	6.767	module=	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	model=	pool=	2.262
-02-19	020.938	3.589	-6.250	16.760	36.177	3.000	4.000	1.000	0.000	0.000	0.000	1.143	280	36.000	15.252	9.735	39.634	47.714	39.634	23.200	module=	32.300	32.300	32.300	32.300	32.300	32.300	32.300	32.300	32.300	32.300	model=	pool=	1.149
-02-05	049.528	5.428	-5.714	15.000	12.754	2.000	3.000	1.000	0.000	0.000	0.000	-1.387	236	-118.000	23.179	22.947	2.579	-0.594	2.579	3.111	module=	15.600	15.600	15.600	15.600	15.600	15.600	15.600	15.600	15.600	15.600	model=	pool=	0.570
-03-03	031.714	20.724	37.271	11.214	62.032	5.000	3.000	1.000	0.000	0.000	0.000	1.248	324	-88.000	6.767	6.767	6.767	6.767	6.767	6.767	module=	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	model=	pool=	9.738
-04-14	014.383	13.183	20.545	4.397	28.493	5.000	5.000	1.000	0.000	0.000	0.000	0.421	525	-82.000	6.099	6.099	6.099	6.099	6.099	6.099	module=	6.099	6.099	6.099	6.099	6.099	6.099	6.099	6.099	6.099	6.099	model=	reg|cal=	8.575
-03-17	023.144	-8.498	15.581	5.461	28.262	5.000	4.000	1.000	0.000	0.000	0.000	2.786	393	90.000	6.767	6.767	6.767	6.767	6.767	6.767	module=	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	model=	pool=	8.262
-03-18	093.021	12.425	8.306	3.952	25.826	3.000	4.000	1.000	0.000	0.000	0.000	-1.559	428	90.000	6.767	6.767	6.767	6.767	6.767	6.767	module=	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	model=	pool=	8.262
-04-02	018.623	23.156	26.200	6.569	35.710	6.000	6.000	1.000	0.000	-1.067	-1.288	-1.087	494	90.000	6.767	6.767	6.767	6.767	6.767	6.767	module=	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	model=	pool=	8.262
-06-12	017.386	12.555	17.756	5.745	33.208	11.000	11.000	1.000	0.000	-10.036	-9.909	0.112	847	-84.000	5.740	5.740	5.740	5.740	5.740	5.740	module=	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	model=	pool=	8.247
-06-12	020.894	-3.715	13.890	3.757	49.379	11.000	9.000	0.000	1.000	27.947	28.494	-0.310	861	-84.000	5.740	5.740	5.740	5.740	5.740	5.740	module=	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	model=	pool=	8.247
-02-19	054.201	5.080	-3.786	1.085	57.720	3.000	3.000	1.000	0.000	0.000	0.000	0.567	275	78.000	-7.403	9.253	-0.357	-3.902	-0.357	0.664	module=	-5.357	-5.357	-5.357	-5.357	-5.357	-5.357	-5.357	-5.357	-5.357	-5.357	model=	reg|cal=	8.161
-05-30	097.704	8.703	6.800	6.047	23.351	9.000	9.000	1.000	0.000	-3.733	-3.795	-0.087	817	92.000	5.740	5.740	5.740	5.740	5.740	5.740	module=	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	model=	pool=	7.753
-04-16	078.114	1.300	-0.333	1.988	56.254	6.000	6.000	0.000	1.000	0.667	0.666	0.324	542	-80.000	6.099	6.099	6.099	6.099	6.099	6.099	module=	6.099	6.099	6.099	6.099	6.099	6.099	6.099	6.099	6.099	6.099	model=	reg|cal=	7.626
-01-24	094.906	14.124	7.107	5.000	7.000	1.000	2.000	0.000	1.000	0.000	0.000	-1.690	159	-60.000	-1.340	7.967	5.261	9.816	5.261	3.156	module=	7.256	7.256	7.256	7.256	7.256	7.256	7.256	7.256	7.256	7.256	model=	pool=	7.600

11. Residual diagnostics plots



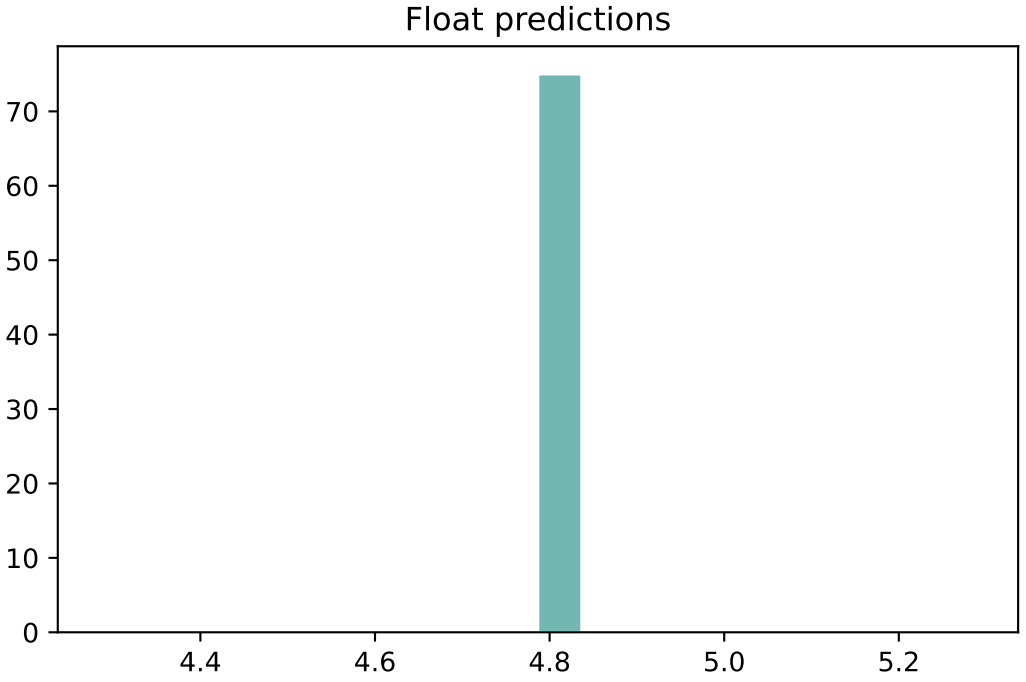
12-14. Final Decisions / Limitations / Next Steps

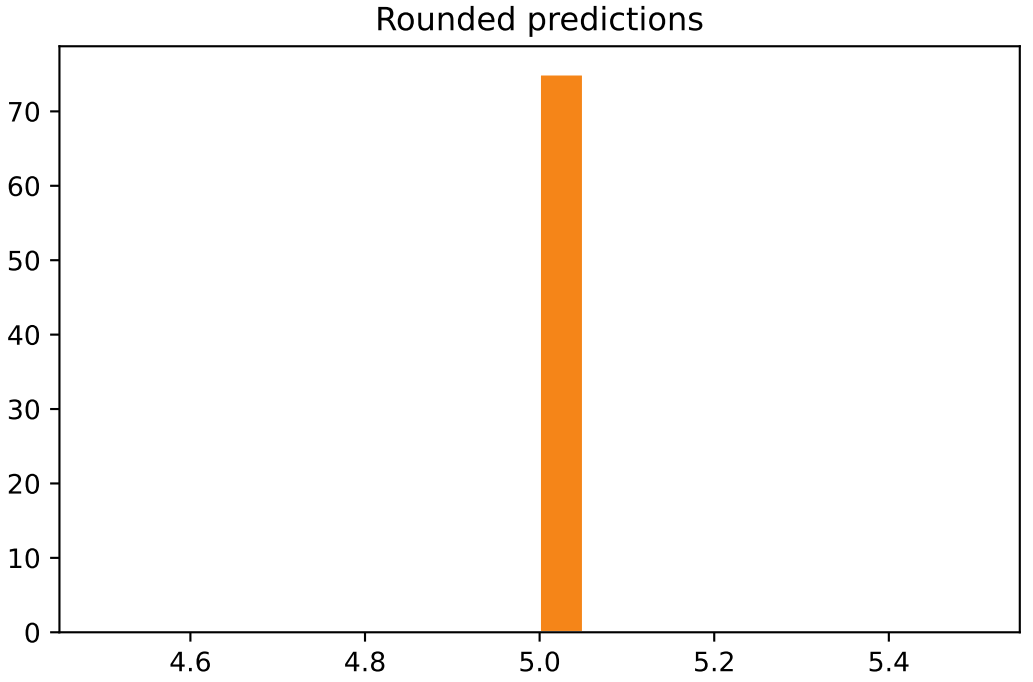
```
12. Final derby prediction distribution
- count=75, mean=4.787, std=0.000
- min=4.787, max=4.787
- quantiles 1/5/25/50/75/95/99 = 4.787, 4.787, 4.787, 4.787, 4.787, 4.787, 4.787

13. Final decisions and rationale
- Final core candidate: elo_base_static|compact_recency|hl=none|ridge=2|hgb=0
- Final postprocess candidate: module=expand_affine_global|stack=pool|cal=pooled|scale=none|q50=0.00|win=0.000|eta=0.55
- Recency profile=compact_recency | half_life=None
- Shift mitigation applied=True
- Dispersion guard applied=False

14. Limitations + next steps
- Derby labels hidden: final selection relies on nested train-only proxies.
- No injuries/lineups/travel features available.
- Future work: monotone piecewise regime calibration and richer Bayesian uncertainty.
```

12. Final derby prediction distribution





Appendix A. First 10 rows of predictions.csv (1/1)

GameID	Date	Team1_Conf	Team1_ID	Team1	Team2_Conf	Team2_ID	Team2	Team1_WinMargin
1941	7/4/2025	Purple	68	Idaho	Yellow	119	Houston	5
1942	7/4/2025	White	105	Helena	Yellow	120	Iowa	5
1943	7/4/2025	Red	82	Jackson	Purple	67	Concord	5
1944	7/4/2025	White	108	Montpelier	Yellow	129	West Virginia	5
1945	7/4/2025	Yellow	121	Lansing	White	113	South Dakota	5
1946	7/4/2025	Crimson	21	Lincoln	Yellow	118	Florida	5
1947	7/4/2025	Yellow	117	Columbia	Crimson	23	Nashville	5
1948	7/4/2025	Orange	63	New Jersey	Yellow	126	Pierre	5
1949	7/4/2025	Purple	77	Wilmington	Orange	58	Honolulu	5
1950	7/4/2025	Crimson	22	Michigan	Green	49	Natchez	5

Appendix B. Top 20 rankings by Rank (1/1)

TeamID	Team	Rank
36	Massachusetts	1
90	Big Sur	2
94	Kentucky	3
37	Minnesota	4
40	Pennsylvania	5
45	Boise City	6
2	Baton Rouge	7
153	Tokyo	8
26	Phoenix	9
92	Illinois	10
88	Port St. Lucie	11
137	Milan	12
46	Boston	13
54	Boise	14
47	Chicago	15
19	Georgia	16
13	Texas	17
85	New Hope	18
131	Belgrade	19
149	Seoul	20

Appendix C. Selected hyperparameters (1/1)

name	value
elo_variant	elo_base_static
feature_profile	compact_recency
half_life_days	none
ridge_alpha	2.0
histgb_idx	0
histgb_params	{'learning_rate': 0.04, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10, 'l2_regularization': 0.3, 'ma
postprocess	module=expand_affine_global|stack=pool|cal=pooled|scale=none|q50=0.00|win=0.000|eta=0.55
postprocess_module	expand_affine_global
postprocess_eta	0.55
winsor_bounds	(-116.61, 113.305)
dispersion_guard_factor	1.0
seed	23
OMP_NUM_THREADS	
MKL_NUM_THREADS	

Appendix D. Final derby stage outputs (head) (1/1)

Date	elo_diff	mass	eff	coef_vol	volatility	volatility_adj	mass_eq	planes_eq	layer_eq	consistency	form	form_adj	class	consistency_r	row_index	pred_rich	pred_hold	pred_hist	pred_rich	pred_hist	pred_q5	pred_q2	pred_q8	stage_rank	stage_rank	stage_cash	stage_spread	stage_spread_fr	spread_q
-07-04 0	-4.319	-0.364	-3.797	35.596	29.274	11.000	12.000	0.000	1.000	11.839	11.198	-0.229			0	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	22.378	14.106	12.458	39.164	40.902	10.000	12.000	0.000	1.000	-20.800	21.102	0.499			1	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	61.146	-3.146	-7.864	59.498	27.316	11.000	13.000	0.000	1.000	9.849	10.915	-0.544			2	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	27.445	2.010	-4.412	44.919	28.756	10.000	11.000	0.000	1.000	21.600	19.898	0.109			3	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	20.059	10.874	16.750	38.940	36.984	12.000	10.000	0.000	1.000	-5.000	-4.746	-0.546			4	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	21.386	8.446	1.765	30.409	33.738	11.000	11.000	0.000	1.000	-21.527	21.221	0.054			5	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	35.060	1.343	2.118	40.055	39.435	11.000	11.000	0.000	1.000	-2.073	-3.274	-0.082			6	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	10.828	14.158	8.985	35.920	32.824	10.000	11.000	0.000	1.000	7.418	7.978	0.381			7	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	34.104	0.807	-0.588	41.956	41.415	11.000	11.000	0.000	1.000	-17.097	17.566	-0.023			8	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	66.423	13.143	21.640	28.163	37.988	10.000	11.000	0.000	1.000	12.927	14.346	1.020			9	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	35.309	6.094	1.544	41.287	34.893	10.000	11.000	0.000	1.000	-3.855	-3.785	0.003			10	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	11.625	-4.250	-8.824	32.105	38.441	11.000	11.000	0.000	1.000	-9.964	-9.016	-0.303			11	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	18.201	1.607	0.608	31.302	40.159	11.000	12.000	0.000	1.000	-0.827	-0.219	0.101			12	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	11.916	-2.432	-8.588	34.862	33.405	12.000	11.000	0.000	1.000	-12.909	12.470	-0.039			13	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	42.745	7.018	9.059	33.643	31.326	11.000	11.000	0.000	1.000	1.200	-0.538	0.401			14	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	26.821	0.733	-3.882	33.735	36.430	11.000	11.000	0.000	1.000	19.600	20.081	-0.155			15	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	10.577	5.344	-0.026	28.877	37.659	12.000	11.000	0.000	1.000	11.533	12.246	0.065			16	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	19.333	-0.458	-2.706	58.962	38.414	11.000	12.000	0.000	1.000	41.327	40.959	-0.193			17	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	31.223	6.648	-6.941	33.206	35.506	11.000	11.000	0.000	1.000	2.327	3.044	-0.150			18	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000
-07-04 0	7.150	-3.761	-6.333	44.267	37.696	12.000	12.000	0.000	1.000	35.100	33.916	-0.273			19	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	10.077	10.077	4.787	0.000